

Private Credit Portfolio — Stress Testing Framework

Built from first principles: three engines, a step-by-step worked example, the treatment of carry, and a structural method for emerging risk

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Companion documents: *SWES Private Credit Framework v1.0* (regulatory scenario application), *SWES APT Shock File*

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The one-page answer

1. A factor model alone will systematically under-read this portfolio. Direct lending loans are floating-rate, so a rate shock produces almost no price move — while raising the borrower's cash interest bill by a quarter to a half. The instrument is rate-insensitive; the *credit* is rate-critical. Section 5 Step 4 shows a rate shock alone breaching a covenant on a loan whose mark barely moves.

2. The marks you would feed a regression are your own opinion, not a market price. Appraisal smoothing halves estimated betas — Section 12.1 derives the exact attenuation factor. Betas must be assigned analytically from spread duration, or estimated on liquid proxies and transplanted.

3. Loss is a threshold process. A linear beta fitted over a sample containing no defaults is a precise estimate of the wrong quantity. You need a fundamental engine alongside the market engine, and the gap between the two is the most informative number in the pack.

4. Carry must be counted over a 6–12 month horizon, but never netted into the headline. Section 6. Gross carry of ~11–12% is the same order of magnitude as the mark impact, so excluding it makes a one-year stress incoherent — and including it carelessly lets a severe recession print a positive total return. The rule: report both views, risk-adjust the accrual for non-accrual timing, split cash from PIK, and make the decision metric **carry minus expected loss**, not carry.

5. On emerging risk, time-series regression will not work — and your segmental-revenue idea is the right answer. Section 13 builds it into a full method: borrower revenue segments → occupation mix → published task-level AI exposure research (Anthropic Economic Index, OpenAI GDPval, Eloundou et al.) → revenue-at-risk → the standard credit cascade. Section 14 runs it end to end and produces the finding that matters: **a structural technology scenario with no macro shock at all does similar EBITDA damage to a 1-in-25 stagflation, but is materially worse for the loan**, because it impairs the exit rather than the cash flow — and direct lending is repaid at the exit.

Part I — Foundations

1. What a private credit loan actually is

Everything in this framework follows from writing down the payoff. A senior secured loan pays the lender:

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Payoff at maturity = min( Face + accrued , Enterprise Value - claims ranking ahead )

Equivalently:      = Face - max( 0 , Face - EV_available )
                   = a risk-free bond MINUS a short put on the borrower's enterprise value,
                                     struck at the debt through our tranche
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Four consequences follow directly, and they drive every design choice in this document:

#	Consequence of the payoff	What it implies for stress testing
1	Upside is capped, downside is not. The best case is par plus coupon.	The distribution is severely left-skewed. A mean or a standard deviation tells you almost nothing. Stress testing is not a refinement of the risk report for this asset class — it <i>is</i> the risk report.
2	Sensitivity to enterprise value is non-linear. $\partial \text{Payoff} / \partial \text{EV} \approx 0$ when EV is comfortably above debt, and $\rightarrow 1$ as EV approaches debt.	A linear factor sensitivity estimated in normal conditions measures the flat region. It cannot generate the loss, because the loss lives in a region the estimation sample never visited. This is the fundamental reason regression fails here (Section 12.1 iv).
3	The value has two independent drivers: the discount rate applied to a performing loan (a <i>market</i> variable), and the probability and severity of ending up on the wrong side of the strike (a <i>credit</i> variable).	Two engines are not a preference; they are what the payoff function requires. Engine A prices the discount rate; Engine B prices the option. Section 3.
4	The strike is set by the borrower's balance sheet; the moneyness moves with EBITDA and the exit multiple.	The stress variables that matter most are EBITDA and the EV multiple — neither of which is a market factor, and neither of which any factor model contains.

The coupon is *compensation* for having written that put. So the honest test of whether a loan is performing acceptably under stress is not "is the coupon positive" but "**does the coupon exceed the expected cost of the option I have written**" — carry minus expected loss. Section 6 makes this the decision metric.

2. Five structural facts that shape the design

Fact	Consequence	Design response
F1. The loans are floating-rate. Rate duration $\approx$ time to next reset (0.08–0.25y). Spread duration $\approx$ expected life (2.0–3.5y).	Price sensitivity to the risk-free curve is near zero <i>by construction</i> : when rates rise, so does the coupon, and the discounted value is unchanged. But the borrower's cash interest bill rises one-for-one. The risk does not disappear — it migrates from the mark to the borrower.	Rate shocks are routed to Engine B as a coupon input. Never report a rate scenario from the factor model alone. See Section 5 Step 4.
F2. The price is an internal valuation , Level 3, committee-approved, and — as the FSB noted in May 2026 — subject to incentives to smooth.	The return series a factor model regresses on is an output of your valuation <i>policy</i> . Regression on it measures marking behaviour. Autocorrelation ϕ of 0.4–0.6 is typical, which halves every estimated beta.	Exposures assigned analytically. A standing mark-to-comparable gap test quantifies the latent lag. Section 12.1.
F3. Loss is a threshold process. Covenant breach $\rightarrow$ waiver $\rightarrow$ equity cure $\rightarrow$ PIK toggle $\rightarrow$ non-accrual $\rightarrow$ restructuring.	Nothing happens, then everything does. And the thresholds have <i>cliffs</i> inside them: equity cure rights are typically limited to two in any four quarters and five over the life. Once exhausted, the next breach is an event of default.	Explicit covenant testing per name; cure-right consumption tracked as a state variable; structural distance-to-default where the linear view is unconvincing.
F4. The book has tens of names, not thousands , with sponsor concentration and heavy clustering in software, business services and healthcare services.	No law of large numbers. Sponsor is an unmodelled common factor: one playbook, one documentation template, one restructuring behaviour across many names.	Granularity adjustment on the loss distribution; explicit jump-to-default and sponsor-cluster scenarios; concentration reported alongside every scenario result.
F5. Repayment comes from refinancing or sale, not from amortisation. Scheduled amortisation is typically 1% p.a.; 95%+ of principal is repaid at exit.	The loan's real dependency is on the <i>exit market</i> being open at maturity at an acceptable multiple. A scenario that impairs terminal value permanently is worse than one that impairs cash flow temporarily, even if the EBITDA damage is identical.	Exit multiple and refinancing availability are first-class stress variables. This is why the AI scenario in Section 14 is worse than the stagflation scenario despite similar EBITDA damage.

3. Target architecture

	Engine A — Market / valuation	Engine B — Fundamental credit	Engine C — Portfolio distribution
Prices	The discount rate on a performing loan	The short put — moneyness and severity	The joint distribution across names
Question	If spreads and curves move, what is the mark?	Will the borrower pay, and if not what do we recover?	How bad can the aggregate get, and what drives the tail?
Tool	FIS APT, or Excel: spread duration $\times$ Δ DM	Excel, per borrower	Excel closed-form (Vasicek) or Monte Carlo
Output	Δ NAV by factor, spread DV01, concentrations	ICR / leverage / LTV paths, breach list, migration, ECL, stressed marks	Expected and unexpected loss, 95/99/99.5th percentiles, contribution by name and sector
Frequency	Monthly sensitivities, quarterly scenarios	Quarterly — top 30 names individually, tail by cohort	Semi-annual and on material portfolio change
Blind to	Defaults; rate-driven credit stress; anything without a factor	Market repricing; relative value; contagion	Everything except its correlation assumption

Lens D — Liquidity and financing sits across all three: NAV facility LTV, subscription line, unfunded commitments and DDTLs, PIK-driven cash yield erosion, and redemption dynamics for evergreen vehicles. Section 15.3.

The reconciliation rule. Engine A's spread widening already prices expected loss on a still-performing loan; Engine B computes realised loss directly. Adding them double-counts. The convention:

A name is in EXACTLY ONE of two states in each period:

PERFORMING → carried in Engine A at the scenario DM + its migration add-on.
Its ECL is a MEMO item. It is NOT added to the loss.

DEFAULTED → removed from Engine A from the period of default.
Loss = prior mark - recovery value. Its spread is irrelevant; it no longer has one.

Portfolio loss = $\sum_{\text{performing}} (1-d) \cdot |\Delta \text{Mark}|$ + $\sum_{\text{defaulted}} d \cdot (\text{prior mark} - \text{recovery})$ - carry

4. The formula set, derived

Every formula below is stated with the one-line reason it is true, so that it can be reproduced or challenged without reference to a model.

4.1 Engine A — the mark

[A1] Value of a floating-rate loan
The coupon resets to the reference rate, so the ONLY thing that can move the price is a change in the credit spread demanded (the discount margin, DM) — plus a small stub to the next reset.

$$\Delta V/V \approx -SD \times \Delta DM - RD \times \Delta r_{ref} \quad [+ \text{carry term over a horizon, Section 6}]$$

SD = spread duration $\approx$ Macaulay duration to EXPECTED repayment (2.0–3.5y typical)
RD = rate duration $\approx$ time to next coupon reset (0.08–0.25y typical)

Because RD is $\sim 20\times$ smaller than SD, the rate term is negligible. This is F1, in one line.

[A2] Stress amplifier — WAL extension
Under stress prepayments stop and amend-and-extend replaces refinancing, so expected life LENGTHENS exactly when spreads widen. Extend EXPECTED LIFE by 1 year in every stress scenario; because duration is shorter than maturity for a coupon instrument, that raises spread duration by roughly 0.75y.
Effect: SD 2.75 $\rightarrow$ 3.50 — a 27% amplification of the same spread shock.

[A3] Total DM shock — two additive, non-overlapping components
ADM_{total} = ADM_{generic} (what the market charges ALL loans of this rating)
+ ADM_{migration} (what it charges for the notches THIS name has fallen)

They do not double-count because the migration leg is priced off rating-notch spread DIFFERENTIALS, not off absolute spread levels.

[A4] Implied DM from a mark (to calibrate the starting point)
DM₀ $\approx$ (Par — Mark) / SD + contractual margin

4.2 Engine B — the credit

[B1] Revenue $\rightarrow$ EBITDA, via operating leverage
Fixed costs do not fall with revenue, so EBITDA falls by a MULTIPLE of the revenue fall.
 $\Delta \text{EBITDA}\% = \Delta \text{Revenue}\% \times \text{OL}$ OL = 1 + FixedCost/EBITDA, or fitted from history
Typical: 1.3–1.6 asset-light services, 1.8–2.5 industrials

[B2] EBITDA quality — reverse the add-backs
Reported covenant EBITDA includes synergies, run-rate and pro-forma items that may never arrive.
 $\text{EBITDA}_s = \text{EBITDA}_0 \times (1 + \Delta \text{Revenue}\% \times \text{OL}) - \rho \times \text{Addbacks}$ ρ = reversal fraction

[B3] Cash interest — on GROSS drawn debt, at the stressed reference rate
 $\text{CashInt}_s = \text{GrossDebt} \times [\max(r_{ref}_s, \text{floor}) + \text{margin}_{cash}]$
Note the floor: at low rates it protects you; at high rates it is irrelevant and must not be left switched on in the model, or the stress will be understated.

[B4] The three covenant tests
 $\text{ICR}_s = \text{EBITDA}_s / \text{CashInt}_s$ vs covenant, typically 1.50–2.00×
 $\text{Leverage}_s = \text{NetDebt} / \text{EBITDA}_s$ vs covenant, typically 30–35% cushion
 $\text{FCCR}_s = (\text{EBITDA}_s - \text{capex} - \text{tax}) / (\text{CashInt}_s + \text{amort})$ vs 1.00×
FCCR below 1.00 is the one that matters economically even where there is no covenant: it means the business cannot fund itself and must draw the revolver or PIK.

[B5] Enterprise value and the loan's moneyiness
 $\text{EV}_s = \text{EBITDA}_s \times \text{Multiple}_s$
 $\text{LTV}_s = \text{NetDebt} / \text{EV}_s$ LTV > 100% $\Rightarrow$ the loan is underwater at going-concern value and the sponsor's option is worthless

[B6] Recovery — a distressed sale, not a going-concern sale
 $\text{Recovery}\$ = \max(0, \text{EV}_s \times (1 - d) - c \times \text{EV}_s - \text{SeniorClaims})$
d = distress discount: 20–25% asset-heavy (property, plant, receivables)
35–45% asset-light (the value walks out of the building)

c = administration / restructuring costs, 5-8% of EV
 SeniorClaims = super-senior RCF (assume FULLY DRAWN – it always is by then)

LGD = 1 - Recovery\$ / Exposure Floor LGD at 10% for timing and process uncertainty.

- [B7] Expected credit loss
 $ECL = PD_s \times LGD_s \times EAD$
 PD_s from the migrated internal grade. Use recession-vintage transition matrices, not TTC.
- [B8] Systematic recovery risk – the correction most models omit
 PD and LGD are POSITIVELY correlated: recoveries fall when defaults rise, because everyone is selling the same assets into the same market. Modelling them independently understates the tail by 20-40%. Apply a downturn LGD, not an average LGD, in any stressed scenario.

4.3 Engine C – the portfolio

- [C1] Single-factor (Vasicek) loss at confidence q
 Every borrower's asset value = $\sqrt{\rho} \times (\text{one common factor}) + \sqrt{1-\rho} \times (\text{its own noise})$.
 Default when that falls below $\Phi^{-1}(PD)$. Condition on a bad draw of the common factor:
- $$L(q) = \sum_i w_i \cdot LGD_i \cdot \Phi \left(\frac{\Phi^{-1}(PD_i) + \sqrt{\rho} \cdot \Phi^{-1}(q)}{\sqrt{1-\rho}} \right)$$
- $\rho \approx 0.18-0.24$ for mid-market corporates. ρ IS the answer – run 0.15 / 0.20 / 0.30 always.
- [C2] Granularity adjustment – because you have 45 names, not 4,500
 Vasicek assumes infinite granularity. With N effective names the true tail is fatter:
 add approximately $(1/N_{\text{eff}}) \times \sum w_i^2 \cdot LGD_i^2 \cdot PD_i(1-PD_i)$ scaled to the quantile.
 $N_{\text{eff}} = 1 / \sum w_i^2$ (the inverse Herfindahl). Report N_{eff} ; it is usually far below the name count.
- [C3] Cohort roll-up – the manual version, which is what you will actually use
 $Loss\% = \sum_{\text{cohorts}} w_c \times [d_c \times (1 - Rec_c) + (1 - d_c) \times |\Delta Mark_c|]$
 This is the reconciliation rule of Section 3 written as arithmetic. See Section 7.

Part II — The worked example

5. Step-by-step borrower stress

5.1 The borrower — all inputs, so the example is fully reproducible

Input	Value	Source / note
Borrower	—	Northwind Business Services Ltd — sponsor-backed mid-market outsourced business services (BPO, IT service desk, managed infrastructure, field services, regulated compliance)
Revenue, R_0	250.0	Segmental split in Section 14 — the same borrower is used for the AI scenario
Adjusted EBITDA, E_0	50.0	20.0% margin
of which add-backs	7.5	15% of adjusted EBITDA — synergies and run-rate. Reported EBITDA is 42.5
Gross drawn debt	250.0	Unitranche, first lien, single tranche
Net debt	250.0	Cash assumed nil for clarity
Super-senior RCF	15.0	Undrawn at T_0 ; ranks ahead of us on enforcement
Opening leverage	5.00×	250 / 50
Margin (cash)	+550bp	No PIK component at origination; PIK toggle available
Reference rate, r_0	4.00%	3-month SOFR; floor 1.00% (out of the money)
All-in coupon	9.50%	4.00 + 5.50
Expected repayment	3.0y	Legal maturity 5.0y — use expected, per [A1]
Spread duration, SD	2.75	Duration to expected repayment
Entry EV multiple	11.0×	EV = 550; equity cushion 300 = 54.5% of EV
Capex / tax / amortisation	5.0 / 3.0 / 2.5	Amortisation 1% p.a. of facility
Covenants	—	Net leverage $\leq 6.50\times$ (30% EBITDA cushion) · ICR $\geq 1.50\times$ · equity cure: 2 in any 4 quarters, 5 over life
Our exposure	40.0	4.0% of a 1,000 fund NAV; current mark 99.0
Operating leverage, OL	2.00×	Fitted from the borrower's 2019–2024 revenue/EBITDA history

5.2 The scenario — S3, stagflationary recession (~1-in-25)

Reference rate **+250bp** · revenue **–10%** · EV multiples **–30%** · add-back reversal **50%** · WAL extension **+1 year** · generic DM widening for B-rated direct lending **+300bp**. Horizon: 12 months.

5.3 The twelve steps

Step	What we are computing	Formula	Substitution	Result	Test
1	Stressed revenue	$R_s = R_0 \times (1 + \Delta Rev\%)$	$250.0 \times (1 - 0.10)$	225.0	—
2	EBITDA after operating leverage [B1]	$\Delta E\% = \Delta Rev\% \times OL$ $E' = E_0 \times (1 + \Delta E\%)$	$\Delta E\% = -10\% \times 2.00 = -20\%$ $50.0 \times (1 - 0.20)$	40.0	Margin falls 20.0% → 17.8%
3	EBITDA after add-back reversal [B2]	$E_s = E' - \rho \times Addbacks$	$40.0 - 0.50 \times 7.5$	36.25	Total EBITDA damage -27.5%, of which 7.5pp is quality, not trading
4	Cash interest [B3]	$Int_s = Debt \times (r_s + m)$	$r_s = 4.00 + 2.50 = 6.50\%$ $250.0 \times (6.50\% + 5.50\%)$	30.00 was 23.75	Interest bill +26.3% — from a shock the mark barely registers
5	Interest cover [B4]	$ICR = E_s / Int_s$	$36.25 / 30.00$	1.208x was 2.105x	FAIL — covenant 1.50x; headroom -19.5%
6	Leverage [B4]	$Lev = NetDebt / E_s$	$250.0 / 36.25$	6.90x was 5.00x	FAIL — covenant 6.50x
7	Fixed charge cover [B4]	$FCCR = (E_s - capex - tax) / (Int_s + amort)$	$(36.25 - 5.0 - 0) / (30.00 + 2.5)$	0.962x was 1.600x	Below 1.00 — the business cannot fund itself; draws the RCF
8	Enterprise value and moneyiness [B5]	$EV_s = E_s \times Mult_s$ $LTV = NetDebt / EV_s$	$Mult_s = 11.0 \times 0.70 = 7.70x$ $EV = 36.25 \times 7.70 = 279.1$ $LTV = 250.0 / 279.1$	89.6% was 45.5%	Equity cushion collapses 300 → 29.1 (10.4% of EV)
9	Recovery in default [B6]	$Rec\$ = EV_s(1-d) - c \cdot EV_s - SeniorClaims$	$279.1 \times (1 - 0.40) = 167.5$ less costs $0.08 \times 279.1 = 22.3$ less RCF drawn 15.0 = 130.2 on debt of 250.0	52.1% LGD 47.9%	d = 40%: asset-light. At entry multiples recovery was 100% — all of the LGD comes from the multiple
10	Migration and PD [B7]	Grade from (Lev, ICR, FCCR) PD from recession matrix	B2 → Caa1 (6.9x, 1.21x, <1.0)	PD 12.0% was 3.5%	Two notches. ECL = $12.0\% \times 47.9\% \times 40.0 = 2.30$ (5.75% of exposure) — <i>memo only</i>
11	The mark [A1-A3]	$\Delta DM = generic + migration$ life +1y → SD_s [A2] $\Delta V/V = -SD_s \times \Delta DM$	$\Delta DM = 300 + 400 = 700bp$ $SD_s: 2.75 \rightarrow 3.50$ $-3.50 \times 7.00\%$	-24.50% mark 99.0 → 74.7	Rate leg: $-0.17 \times 2.50\% = -0.42\%$, immediately offset by the higher coupon — negligible, as [A1] predicts
12	Carry over 12 months [Section 6]	Gross = $(r_s + m) \times h$ less non-accrual split cash / PIK	$12.00\% \times 1.0 = 12.00\%$ less $12.0\% \times 0.5 \times 12.00\% = 0.72\%$ = 11.28%, of which 40% PIKs	+11.28% cash 6.77 PIK 4.51	Carry – EL = $11.28\% - 5.75\% = +5.53\%$ — still being paid for the risk

Result, and why it is a fail

Both maintenance covenants breach, and fixed charge cover falls below 1.00x. The sequence that follows is mechanical and should be stated in the credit paper, because each stage has a different cost:

1. **Breach at the first test date.** The sponsor's options are an equity cure or a waiver. A cure of the ICR breach requires roughly 8.0 of fresh equity (enough to lift EBITDA-equivalent to 45.0, or to prepay debt); a leverage cure requires prepayment of about 14.4. The sponsor will compare that to an equity cushion now worth 29.1 against an original 300.
2. **Cure rights are finite.** Two in any four quarters, five over the life. In a multi-year stress this is a cliff: once consumed, the next breach is an event of default with no negotiating room. **Track cure consumption as a portfolio state variable** — it is the difference between a covenant breach that is an inconvenience and one that is terminal.
3. **FCCR < 1.00 means the RCF gets drawn.** That 15.0 ranks ahead of us, and Step 9 already assumes it is fully drawn — which is why our recovery is 52% rather than 58%. **Undrawn super-senior facilities are a recovery leakage, not a liquidity comfort.**
4. **PIK toggle.** At ICR 1.21x the borrower will elect to PIK. Our accrual continues, our cash yield halves, and leverage compounds — every PIK quarter makes the next test worse. This is why Step 12 splits cash from PIK.
5. **The sponsor decision.** At LTV 89.6% the equity is out of the money on any realistic exit. Sponsor support is a function of moneyness, not of relationship. Expect support to weaken sharply as LTV passes ~85%.

Repercussion for us: a mark of 74.7 (–24.5%), a name moved to the watch-list, an internal downgrade of two notches, an expected loss of 5.75% of exposure, and — critically — a name whose repayment now depends on an exit market that the same scenario has closed (F5).

5.4 The two engines, side by side — this is the report

	Engine A alone (factor model)	Engine B alone (fundamental)	Reconciled — what we report
Generic DM widening only	$-300\text{bp} \times 2.75 = -8.25\%$	—	Mark –24.50% , driven 34% by the market and 66% by this name's own deterioration.
WAL extension	$-300\text{bp} \times 0.75 = -2.25\%$	—	
Migration add-on	Not visible — APT has no view on this borrower's EBITDA	$-400\text{bp} \times 3.50 = -14.00\%$	ECL of 5.75% is a memo , not an addition — the name has not defaulted, so per Section 3 the spread prices it. Carry +11.28% shown separately.
Default / recovery	Not visible	ECL 5.75% (memo)	One-year total return: –13.22%

The number that changes the conversation

Run in isolation, the factor model says this loan loses **8.25%**. The full picture is **24.50%**. The factor model is not wrong — it correctly prices a still-performing B-rated loan when the market repricing is +300bp. It is simply blind to the fact that this particular borrower is no longer a B-rated credit. **Two-thirds of the loss is invisible to Engine A, and no improvement in factor coverage or beta estimation would reveal it**, because the missing information is this borrower's income statement.

6. Carry: should it count over a 6–12 month horizon?

Yes — and it must be counted, or a one-year stress is internally incoherent. But the way it is counted determines whether the exercise is informative or misleading, so the rules matter more than the answer.

6.1 Why it cannot simply be ignored

Gross carry on this book is roughly 11–12% a year. The S3 mark impact is roughly 19% at portfolio level. These are the same order of magnitude. A one-year stress that omits carry is not conservative — it is answering a different question (instantaneous economic value) while labelling it with a one-year horizon. Over 12 months the fund genuinely does collect coupon on the performing balance, and pretending otherwise makes the result impossible to reconcile to the actual P&L, which destroys the framework's credibility with the investment team.

6.2 Why it cannot simply be added

Naive inclusion lets a severe recession print a modest or even positive total return, which is exactly the outcome a stress test exists to prevent. Four distinct errors produce that flattery:

Error	Why it is wrong	Correction
Accruing on defaulted names	A loan that goes non-accrual in month 7 pays for six months, not twelve — and the accrual already booked is reversed.	Half-period convention: multiply gross accrual by $(1 - PD \times 0.5)$. Refine to a quarterly default timing profile if the stress is multi-year.
Treating PIK as income	PIK accrual is the borrower issuing you more of its own paper. On a name migrating to CCC, that paper is worth what the loan is worth — not par.	Split cash and PIK carry. Never count PIK in a distribution-coverage or liquidity metric. Consider marking PIK accrual at the loan's own stressed price.
Using a fixed carry across scenarios	Carry is floating and therefore scenario-correlated . In a rate-cutting recession (S2) carry falls just as losses rise. In stagflation (S3) it rises — but the rise is the very thing breaking the borrowers.	Recompute carry from each scenario's own reference-rate path. Never a constant assumption.
Forgetting it is compensation, not profit	The spread was set to cover expected loss. Counting the whole spread as an offset against loss is partly circular.	Report carry minus expected loss as the decision metric. That is the excess return for bearing the risk, and it is what should be positive.

6.3 The rules

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[H1] Horizon determines treatment
Instantaneous shock    → EXCLUDE carry. The question is "what is it worth now".
Up to 6 months         → exclude from headline; show as a memo line (~5-6%).
6 to 12 months         → INCLUDE, as a SEPARATE line in the bridge. Never netted.
Beyond 12 months       → include with a full path: balance decline, defaults by year,
                        redeployment of repayments at scenario spreads, fund costs.

[H2] The carry calculation
Gross accrual          = [ max(r_ref,s , floor) + margin_total ] × h
Non-accrual adj        = × (1 - PD_s × h × 0.5)                half-period convention
Cash carry             = net accrual × (1 - PIK share)
PIK carry              = net accrual × PIK share              shown separately, excluded from cash metrics
Fund-level net         = Σ w_i · cash carry_i - financing cost - management fee

[H3] The decision metric
Excess carry           = net accrual - (PD_s × LGD_s)
Positive ⇒ still being paid for the risk taken.
Negative ⇒ NOT being paid. This is the single most useful line in a stress report,
and it is the line that separates the AI scenario from the macro scenario
in Section 14 — same EBITDA damage, opposite answer.

[H4] Presentation — always both views
Report the MARK view (economic value, no carry) AND the TOTAL RETURN view (with carry).
Risk appetite limits expressed on economic value are NOT offset by carry: you cannot
distribute a mark, and a NAV facility covenant tests the mark, not the return.
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6.4 Applied to Northwind — the same loan, three horizons

Horizon	Mark impact	Gross accrual	Non-accrual adj.	Net carry	of which cash	Total return	Excess carry [H3]
Instantaneous	-24.50%	—	—	—	—	-24.50%	—
6 months	-24.50%	+6.00%	-0.18%	+5.82%	+3.49%	-18.68%	+2.94%
12 months	-24.50%	+12.00%	-0.72%	+11.28%	+6.77%	-13.22%	+5.53%

Non-accrual adjustment at 12m = $12.00\% \times (12.0\% \times 1.0 \times 0.5) = 0.72\%$. Excess carry at 12m = $11.28\% - (12.0\% \times 47.9\%) = +5.53\%$. PIK share assumed 40% once the toggle is elected. The cash column is what actually funds distributions; the difference between -13.22% and -17.73% (mark plus cash carry only) is accrued paper.

How to say this to the Head of Credit in one sentence

"Over twelve months this name loses 24.5% of its mark and earns 11.3% of coupon, of which only 6.8% is cash — so the economic loss is 24.5%, the reported total return is -13.2%, and the honest question is whether an 11.3% coupon is adequate for a 5.75% expected loss. Today it is. In the technology scenario in Section 14, it is not."

7. Portfolio roll-up and fund leverage

Same scenario, applied to a stylised fund: NAV 1,000, gross assets 1,333, NAV facility drawn 333 (LTV 25.0%, covenant 40.0%), 45 borrowers grouped into three cohorts. Per [C3].

Cohort	Weight	1yr default rate d	Recovery	ΔMark, survivors	Default loss $d \times (1 - \text{Rec})$	Mark loss $(1 - d) \times \Delta \text{Mark} $	Cohort loss	Net carry $12\% \times (1 - d/2)$
A Resilient	40%	2%	65%	-7.5%	0.70%	7.35%	8.05%	11.88%
B Core	45%	10%	52%	-20.0%	4.80%	18.00%	22.80%	11.40%
C Watch-list	15%	25%	45%	-30.0%	13.75%	22.50%	36.25%	10.50%
Weighted total (asset level)							18.92%	11.46%

Step	Workings	Mark view	1-year total return view
Gross assets at T_0	$\text{NAV } 1,000 \div (1 - 0.25)$	1,333.0	1,333.0
Asset return	mark: -18.92% total: -18.92% + 11.46%	-18.92%	-7.46%
Assets after	$1,333 \times (1 + \text{return})$	1,080.8	1,233.6
Less NAV facility financing cost	$333 \times 9.00\%$	—	-30.0
Less management fee	$1,333 \times 1.25\%$	—	-16.7
Less facility principal	drawn	-333.0	-333.0
NAV after	—	747.8	853.9
NAV impact	vs 1,000	-25.2%	-14.6%
NAV facility LTV after	$333 \div \text{assets}$	30.8%	27.0%
Covenant test ($\leq 40\%$)	—	PASS	PASS

Reading the two columns

Both numbers are correct and both must be reported. -25.2% is the change in economic value and the number that tests against NAV facility covenants, risk appetite and any capital-style metric. -14.6% is the one-year total return and the number that will appear in the LP report. The 10.6 percentage point difference is carry — and roughly 40% of that carry is PIK, so the cash-collected figure is lower still.

Note the leverage amplification. A 18.92% asset decline becomes a 25.2% NAV decline because the facility does not shrink. That is a 1.33x multiplier — modest here, but it scales linearly with LTV, and it is the reason the fund-level lens (Section 15.3) is not optional.

8. Reverse stress on the same borrower and fund

The same arithmetic, inverted. Instead of assuming a shock and reporting a loss, fix the outcome and solve for the input. This removes the "is your scenario severe enough?" argument entirely, because severity becomes an output.

Question	Solve	Answer	Judgement
What EBITDA breaches the leverage covenant?	$E^* = \text{NetDebt} / 6.50 = 250 / 6.50$	38.46 = -23.1% from 50.0	Equivalent to a revenue fall of 11.5% at OL 2.0 — a normal recession
What EBITDA breaches the ICR covenant at today's rates?	$E^* = 1.50 \times 23.75$	35.63 = -28.8%	Leverage binds first at current rates
What EBITDA breaches ICR if rates rise 250bp?	$E^* = 1.50 \times 30.00$	45.00 = -10.0%	The rate shock moves the breach point from -28.8% to -10.0%. This is the single most important sensitivity in the file
What rate rise alone , with EBITDA flat, breaches ICR?	$\text{Int}^* = 50 / 1.50 = 33.33$ $\text{all-in}^* = 33.33/250 = 13.33\%$ $r^* = 13.33\% - 5.50\% = 7.83\%$ $\text{rise} = 7.83\% - 4.00\%$	+383bp all-in 13.33%	Closer than it looks — and this holds EBITDA <i>flat</i> . The S3 shock is already +250bp of that 383bp
At what exit multiple do we stop being fully covered?	$\text{Mult}^* = \text{NetDebt} / E_s = 250 / 36.25$	6.90x	vs 7.70x in the scenario — 10% of headroom on the multiple
What NAV decline breaches the fund's 40% LTV covenant?	$\text{Assets}^* = 333 / 0.40 = 832.5$ $\text{decline} = 1 - 832.5/1,333$	-37.5% of assets = -50.0% of NAV	Roughly 2x the S3 mark impact — genuine headroom
What (default rate, LGD) pair costs 10% of NAV?	$d \times \text{LGD} = 0.10 \times (\text{NAV}/\text{assets})$ = 0.075 on assets	e.g. 15% × 50% or 10% × 75%	Plot as an iso-loss curve with GFC, COVID and 2015–16 energy overlaid

The third row is worth putting on its own slide. It is the cleanest possible statement of F1: **a 250bp rate rise does not move the mark, and it cuts the borrower's margin for error from 29% to 10%.**

Part III — The approach menu

9. Twenty approaches, graded

Tags show where each is best implemented: **APT** factor model · **XL** manual/Excel · **BOTH** either.

Family A — Market and valuation stress

#	Approach	Mechanics	What it tells you	Effort
A1	Analytic spread-duration shock BOTH	[A1] with a grid of DM shocks (+50/100/200/300/500bp), differentiated by internal rating and sector. SD to expected repayment, with the [A2] extension in every stress.	The workhorse; spread DV01 in % of NAV.	Low
A2	Curve-shape scenarios BOTH	Level $\pm 100/250$ bp, bear flattener, bull steepener. The output that matters is the reference-rate path handed to Engine B, not the price.	Proves F1 explicitly.	Low
A3	Historical replay APT	Realised factor moves: GFC Sep-08→Mar-09, COVID Feb→Mar-20, 2022 rates, Aug-2011, 2015–16 energy, late-2025 loan dislocation.	Severity calibration without scenario invention.	Low
A4	Factor-model beta propagation APT	APT's native route — only with the beta hygiene of Section 12.2.	Cross-asset consistency with the rest of the platform.	Med
A5	Mark-to-comparable gap XL	Reprice the performing book at observable comparable yields (BSL secondary DMs, listed BDC-implied yields, Lincoln / Cliffwater / KBRA marks) and report the gap to carrying value.	Highest value per hour in the framework. Quantifies latent valuation lag before anyone else does.	Low
A6	Implied-DM dispersion XL	[A4]: solve each loan's implied DM from its mark; rank against origination and against the rating-matched index. Flag names whose implied DM has not moved while fundamentals have.	Finds stale marks at name level, not just portfolio level.	Low

Family B — Bottom-up fundamental credit

#	Approach	Mechanics	What it tells you	Effort
B1	Revenue → coverage cascade XL	Section 5 in full: [B1]–[B7], twelve steps.	Who breaks, when, and on which covenant.	Med
B2	EBITDA quality stress XL	[B2]. Reverse 50% and 100% of add-backs as standing sensitivities.	Moves leverage by a turn or more with no macro shock at all — 7.5pp of the 27.5% EBITDA damage in Section 5.	Low
B3	Structural / Merton distance-to-default XL	Asset value = EBITDA × sector multiple; asset vol from listed comparables (25–40%); default point = net debt. $DD = [\ln(V/D) + (\mu - \sigma^2/2)T] / \sigma\sqrt{T}$.	Needs no loan price history and produces the non-linearity directly. The right tool where there is no track record.	Med
B4	Recovery / LGD stress XL	[B6] with the distress discount d as the key parameter, plus [B8] downturn LGD.	LGD is where the last five years have actually disappointed.	Med
B5	Rating migration matrices BOTH	Recession-vintage transition matrices; price migration at rating-notch spread differentials [A3].	Bridges Engine B back into Engine A marks without double counting.	Med
B6	Borrower liquidity runway XL	Months = (cash + <i>available</i> revolver) ÷ monthly burn after interest, capex and working capital. Stress revolver availability — springing covenants remove it exactly when needed.	The true leading indicator. Borrowers default when cash runs out, not when a ratio breaches.	Med
B7	Hedge roll-off XL NEW	Most sponsor deals carry an interest rate cap or collar for 2–3 years, not for the loan's life. Model the borrower's rate exposure <i>after</i> the hedge expires, at the scenario's forward rate and the scenario's cap premium.	Frequently the single largest unmodelled cliff in a mid-market book, and invisible in every ratio until the quarter it happens.	Low
B8	Cure-right exhaustion XL NEW	Track equity cures consumed against the 2-in-4 / 5-over-life limit as a state variable. Model the sponsor's willingness to cure as a function of LTV, not of relationship.	Converts "breached" into "breached with no remedy" — a different credit entirely.	Low
B9	Maturity wall / refinancing XL NEW	Ladder exposures by expected repayment. In stress, assume the exit market is closed: model amend-and-extend economics (fee, margin step-up, extended SD) and the fraction that cannot refinance at all.	F5 made operational. Directly relevant to the AI scenario, where the impairment is permanent.	Med

Family C — Portfolio, concentration and reverse stress

#	Approach	Mechanics	What it tells you	Effort
C1	Vasicek loss distribution XL	[C1] with [C2] granularity adjustment. Always run $p = 0.15 / 0.20 / 0.30$.	Expected vs unexpected loss; a 99.5th percentile for risk appetite.	Med
C2	Multi-factor Monte Carlo BOTH	As C1 plus sector factors <i>and an explicit sponsor factor</i> ; correlated latent variables, 50k paths.	Quantifies sponsor concentration, which no standard model captures.	High
C3	Jump-to-default / cluster XL	Default the top 1/3/5 obligors at downturn LGD; separately default one sponsor's names, or one sector.	Trivial to compute and usually larger than the generic spread scenario.	Low
C4	Reverse stress testing XL	Section 8.	Severity becomes an output, not an argument.	Low
C5	Distance-to-covenant surface XL	Solve the Section 8 row 1–2 calculation for every name; plot the distribution; report exposure-weighted share within 10/20/30% of breach.	The standing metric. One chart, quarterly, showing fragility move before any loss appears.	Low
C6	Vintage cohort analysis XL NEW	Group by origination year. The 2021–22 vintage was underwritten at 6.5–7.0× leverage against 1–2% base rates; the same leverage at 4–6% base rates is a different credit.	Isolates the cohort most likely to fail, without any scenario at all.	Low
C7	PD–LGD correlation XL NEW	[B8]. Re-run C1 with LGD as an increasing function of the common factor rather than a constant.	Typically fattens the 99.5th percentile by 20–40%. Standard in bank credit modelling and routinely omitted in fund stress tests.	Med

Family D — Liquidity, financing and behaviour

#	Approach	Mechanics	What it tells you	Effort
D1	NAV facility LTV stress XL	Section 7. Test the <i>mark</i> view against the covenant; model advance-rate tightening at renewal and the forced-deleveraging path.	The amplifier that turns –18.9% of assets into –25.2% of NAV.	Low
D2	Commitment and DDTL draw XL	Assume stressed borrowers draw 100% of delayed-draw and revolver availability concurrently.	Undrawns convert to funded exposure at the worst moment — and rank ahead of you (Step 9).	Low
D3	Cash yield / PIK erosion XL	[H2]. Migrate a stressed share from cash-pay to PIK; recompute distributable cash and distribution coverage.	Income risk, distinct from NAV risk — and the trigger for redemptions in evergreen vehicles.	Low
D4	Redemption / first-mover XL	Stressed redemption as % of NAV; sources waterfall (cash → liquid sleeve → revolver → secondary sale at discount); gate and notice mechanics.	Where a valuation lag becomes a real transfer of value between LPs.	Med
D5	Second-round effects XL	Forced secondary sales at observed LP-stake discounts (20–35%) feeding back into marks; sponsor support fatigue where equity is underwater.	The difference between a paper loss and a realised one.	Med

10. Completeness matrix and gap register

A stress framework should be able to state what it does *not* cover. Mapping risk type against approach:

Risk	Covered by	Residual gap and treatment
Credit spread / market repricing	A1, A3, A4	Covered.
Interest rate level and shape	A2 → B1, B7	Covered, provided rate shocks are routed to Engine B. Basis risk (term SOFR vs OIS vs the hedge index) is immaterial here and is not modelled.
Borrower earnings deterioration	B1, B2	Covered. Operating leverage is the key judgement — fit it per borrower and disclose it.
Default and recovery	B4, B5, B6, C7	Covered. Recovery data for private credit is thin and drawn from a benign period; treat LGD as the least reliable parameter in the framework.
Concentration (obligor, sector, sponsor)	C2, C3	Covered. Sponsor correlation is assumed, not estimated — there is no dataset for it. State the assumption.
Refinancing / exit market	B9, F5	Covered from v2.0. Extension probability is a judgement input.
Documentation / LME	S8, B4	Covered as an LGD add-on. Not a market factor and never will be — it is a legal review that produces a number.
Vintage / underwriting standards	C6	Covered from v2.0.
Fund leverage and financing	D1, D2, Section 7	Covered.
Liquidity and redemptions	D3, D4, D5	Covered for evergreen vehicles; closed-end funds need only D2.
FX	A4 (APT), hedge cost in D1	Partial gap. Cross-currency mismatch between a borrower's revenue currency and its debt currency is not modelled per-name. Add a flag to the loan tape and a manual overlay for the affected names.
Emerging structural risk (technology, transition)	Sections 13–14	Covered by method, not by data. Carries no probability and must never be presented with one.
Valuation policy / model risk	A5, A6	Partial gap. Add a "valuation lag" scenario: apply last quarter's comparable marks and measure the difference. Cheap, and it quantifies the first-mover exposure the FSB flagged.
Operational, legal, key-person	—	Out of scope by design; handled in the operational risk framework. Stated here so the omission is deliberate rather than accidental.

11. Standing scenario suite

Calibration is **indicative and refreshed at each jump-off** from Bloomberg. Anchors as at mid-2026: direct lending spreads around SOFR+450–650bp, BDC non-accruals around 3.8% of cost with weaker lower-middle-market lenders approaching 5%, PIK around 8% of BDC investment income.

Tier 1 — sensitivities, monthly, fully mechanical

DM +50/100/200/300/500bp · reference rate ±100/±250bp · EBITDA –10/–20/–30% · EV multiple –1/–2/–4 turns · WAL +1y · add-back reversal –50/–100% · LGD +10/+20pts · top-1/3/5 obligor default · hedge roll-off.

Tier 2 — coherent scenarios, quarterly

Scenario	Δ ref rate	Δ DM	Revenue	EV mult.	Default	LGD	Purpose
S1 Mild slowdown (~1-in-5)	−100bp	+100bp	−3%	−10%	4%	40%	Base adverse; risk appetite calibration
S2 Hard landing (~1-in-20)	−250bp	+350bp	−10%	−25%	9%	45%	Classic recession — rate relief partly offsets, and carry falls
S3 Stagflation (~1-in-25)	+250bp	+300bp	−10%	−30%	10%	50%	Worst for this asset class — worked through in Sections 5–8
S4 Credit-only dislocation	0	+250bp	0%	−15%	3%	40%	Technical/flow-driven; tests the mark, not the credit
S5 Higher-for-longer, growth holds	+150bp	+75bp	+2%	−10%	6%	40%	Slow attrition — the most likely path, and the one Engine A scores at ~zero
S6 Sponsor / sector cluster	0	+150 / +600*	−20%*	−35%*	25%*	55%	Concentration; * affected cohort only
S7 Technology displacement	0	+200bp*	−17%* net	−40%*	15%*	61%	Sections 13–14. Structural, not cyclical
S8 Documentation / LME	0	+100bp	−5%	−15%	7%	65%*	Recovery-only stress on weakly documented names

The deliberate contrast is S2 against S3. Both are severe recessions, but in S2 falling reference rates cut the borrower's interest bill just as EBITDA falls — and cut your carry. In S3 they rise. Running both is what demonstrates that this book's dominant vulnerability is the *co-movement* of rates and earnings, not the level of either.

Part IV — Emerging risk

12. Why regression fails, from first principles

Your question was whether time-series regression will give accurate results for emerging risks. It will not, and the reasons are worth deriving rather than asserting, because the same derivation shows exactly where regression *is* reliable.

12.1 The four failure modes

(i) Smoothing attenuation — derivable, and testable in ten minutes

Suppose the true return is r_t but the valuation committee only partially incorporates new information each period, so the reported return is a weighted average of the truth and last period's report: $r_{\text{obs},t} = (1-\phi) \cdot r_t + \phi \cdot r_{\text{obs},t-1}$. Regressing r_{obs} on a factor f gives:

$$\begin{aligned}\beta_{\text{obs}} &= \text{Cov}(r_{\text{obs}}, f) / \text{Var}(f) \\ &= \beta_{\text{true}} \times (1 - \phi) / (1 - \phi \cdot \rho_f) \quad \rho_f = \text{autocorrelation of the factor}\end{aligned}$$

$$\begin{aligned}\phi = 0.5, \rho_f \approx 0 &\rightarrow \beta_{\text{obs}} \approx 0.50 \times \beta_{\text{true}} && \text{you halve your risk} \\ \phi = 0.6, \rho_f \approx 0.1 &\rightarrow \beta_{\text{obs}} \approx 0.42 \times \beta_{\text{true}} && \text{you take out 58\% of it}\end{aligned}$$

Diagnostic: run an AR(1) on your reval series. ϕ above ~ 0.25 confirms smoothing; private credit typically prints 0.4–0.6.

Fixes, in order of preference: (1) do not regress — assign spread duration analytically; (2) estimate on a liquid proxy matched on rating, sector and leverage, and transplant; (3) Dimson summed lags, $\beta = \sum_{k=0..3} \beta_k$; (4) Geltner unsmoothing, $r_t^* = (r_t - \phi r_{t-1}) / (1 - \phi)$. Validate the result against the expected 0.6–0.9 range versus the leveraged loan index.

(ii) Sample poverty and collinearity

Weekly data over three years is 156 observations, but smoothing means the effective independent sample is a fraction of that, and credit factors are mutually correlated at 0.7–0.9. Fit five and you get nonsensical signs and betas that flip when the window moves a quarter. **Rules:** at most two or three factors per sleeve; estimate at bucket level (rating $\times$ sector), never loan level; impose sign constraints; report a ± 1 standard error band and run the stress at both bounds.

(iii) Regime dependence

Credit betas estimated in calm conditions understate crisis co-movement by roughly 1.5–2.5 $\times$. **Fix:** compute a stress beta peak-to-peak over crisis windows and prefer it wherever it exceeds the OLS estimate; or a threshold model conditioning on HY OAS above $\sim 500\text{bp}$; or quantile regression at the 5th–10th percentile. In a stress framework, the calm-period OLS beta is a **lower bound**, never a central estimate.

(iv) Non-linearity — the one no estimator can fix

From Section 1: the loan is a risk-free bond minus a short put on enterprise value. $\partial V / \partial EV$ is near zero when EV is $2\times$ debt and approaches 1 as EV approaches debt. A regression fitted over a sample in which no borrower defaulted estimates the flat region of that curve. It is not a noisy estimate of the tail sensitivity — **it is a precise estimate of a different quantity**. Section 5 shows this concretely: two-thirds of the loss came from a migration that no factor exposure could have predicted, because the information required was Northwind's income statement.

12.2 Where regression is the right tool

Use case	Verdict	Conditions
Mapping one liquid factor to another — e.g. deriving a direct-lending DM path from the BSL loan spread and single-B OAS (the satellite expansion in the SWES framework)	Yes — primary tool	Both series are traded prices; ≥ 10 years spanning a crisis; ≤ 3 anchors; cross-checked against a crisis-window stress beta; prudence margin where $R^2 < 0.5$.
Sector or rating tilts within a liquid credit universe	Yes, with care	Bucket level, liquid proxies, sign-constrained.
Your own portfolio's beta from its own NAV history	No	Fails (i) and (iv) simultaneously. Use only as a smoothing diagnostic.
Loan-level betas inside APT from reval history	No	Fails (i), (ii) and (iv). Assign analytically via FRN terms.
Any factor without a crisis in sample	No	Use a stress beta or judgement.
Emerging risks	No	Section 12.3.

12.3 Emerging risk — three distinct problems

Problem	Why regression fails	What to do instead
(a) No factor history. The driver has no time series.	The regression is not biased; it is undefined.	Section 13's segmental method. Synthetic thematic basket only for severity calibration.
(b) Structural break. The factor exists but the relationship has changed. Pre-2022, higher rates were mildly <i>good</i> for a floating-rate book. Post-2022 the sign inverted as coverage broke.	Regression estimates the pre-break world and reports it confidently, because the pre-break data are plentiful and the post-break data are not.	Rolling-beta plot and a Chow/CUSUM break test as a standing diagnostic; estimate on the post-break sub-sample and widen the band deliberately.
(c) A new common factor emerges. Risks that were idiosyncratic become correlated — one sponsor's playbook, one documentation feature, one shared end-market.	The factor model's core assumption is that residuals are uncorrelated. That assumption breaks precisely when the emerging risk crystallises, and the model reports diversification that no longer exists.	Monitor average pairwise residual correlation — see below.

Standing diagnostic: residual correlation drift

Take the APT residuals — the part of each position's return the factor model does not explain — and plot the average pairwise correlation across the book on a rolling 26-week window. Under the model's own assumptions it should sit near zero and stay there.

When it rises, either a common driver has appeared that your factor set does not contain, or your marks are moving together for a policy reason. Both are things you want to know early, and neither appears in any standard risk report. Run the same test on the fundamental side — rolling correlation of quarterly EBITDA surprises across borrowers — where it needs no market data at all.

13. The segmental-revenue method for emerging risk

Your instinct is the right one, and it generalises into a complete method. The insight is that **you do not need a beta if you have an exposure**. A factor model asks "how has this asset historically co-moved with X?" — unanswerable for a new X. The segmental approach asks "how much of this borrower's revenue is doing the thing that X replaces?" — which is answerable today, from the borrower's own management accounts, with no history at all.

13.1 The chain

Borrower revenue by segment	← you already have this in the credit file / MI pack
↓ map segment to service line and NAICS/SIC	
Occupation mix of that service line	← BLS OES publishes occupation mix by industry
↓ map occupations to tasks	
Task-level AI exposure	← Anthropic Economic Index (O*NET task mapping, automation vs augmentation split); OpenAI GDPval (model performance on real deliverables, 44 occupations across 9 GDP sectors); Eloundou et al. occupational exposure scores; Felten AIOE
↓ apply the three channels below	
Revenue at risk, margin at risk	
↓ timing gate: contract duration vs LOAN maturity	
Effective revenue decline within the loan's life	
↓ [Bl] operating leverage, less cost-side deflation offset	
ΔEBITDA → the standard Section 5 cascade → ICR, leverage, LTV, LGD, mark	

13.2 Three channels, and why the sign is not obvious

The most common analytical error is to treat "AI exposure" as uniformly negative. For a services borrower there are three distinct effects and they do not point the same way:

Channel	Mechanism	Effect on borrower	Dominates when
1. Revenue substitution (demand side)	The client stops buying the service because the model does it. The borrower's revenue is the client's labour cost.	Revenue ↓↓	Labour-arbitrage businesses: BPO, transactional processing, L1 support, staffing.
2. Price deflation	Volume holds but the price falls toward the new cost floor.	Revenue ↓, margin ~flat	Competitive, undifferentiated delivery with annual repricing.
3. Cost deflation (supply side)	The borrower's own delivery cost falls; it keeps some of the saving.	Margin ↑	Fixed-price or outcome-based contracts, where the borrower owns the productivity gain.

Net effect = channel 1 + channel 2 + channel 3. For labour-arbitrage businesses channel 1 dominates and the net is sharply negative. For fixed-price managed services, channel 3 can dominate and the net is *positive*. Getting this right per segment is the whole exercise, and it is why a single sector-level haircut is the wrong tool.

13.3 The free screen — run this before any research

Two questions discriminate better than sector classification and cost nothing:

Screen	Exposed	Insulated
Pricing basis — is revenue indexed to headcount?	Per-FTE, per-seat, per-ticket, per-transaction, time-and-materials. Revenue is literally a count of human hours.	Outcome-based, per-asset, per-visit, regulated tariff, consumption of a physical resource.
Contract duration vs loan maturity	Annual renewal, or contracts repricing well before the loan matures.	Multi-year contracted with volume commitments running past the loan's expected repayment.

The insight that makes this a credit question rather than a strategy question

The relevant question is not "is this borrower AI-exposed?" but "**does the displacement arrive before my maturity?**" A borrower whose contracts run three years and whose loan repays in two has an equity problem, not a credit problem — the sponsor absorbs it. The same borrower with a four-year loan has a credit problem. This timing gate is the difference between an interesting observation and a provision, and no sector-level analysis can produce it because it depends on *your* loan's tenor.

13.4 Using the published research honestly

Source	What it gives you	How to use it — and its limits
Anthropic Economic Index	Real Claude conversations mapped to O*NET tasks across ~20,000 task statements, with an automation-versus-augmentation split by occupation. Published with open datasets.	Best available evidence on <i>which tasks are actually being done by AI today</i> , as opposed to which could be. Use the automation share as the numerator for channel 1. Limit: it measures current adoption on one platform, so it understates the three-year end state and reflects that platform's user mix.
OpenAI GDPval	Model performance on real occupational deliverables across 44 occupations in the nine largest GDP-contributing US sectors, graded by expert pairwise comparison.	Best available evidence on <i>capability</i> — whether the work can be done to professional standard. Use it to set the ceiling on displacement. Limit: capability is not deployment; organisational and regulatory friction sits between them.
Eloundou et al. exposure scores; Felten AIOE	Occupation-level exposure indices derived from O*NET task content.	Convenient numeric scores for a first pass across many occupations. Limit: exposure is not impact, and these predate current model capability.
BLS OES	Occupation mix by industry — the join that connects a revenue segment to a set of occupations.	The essential linking table. Limit: US industry averages, not your borrower's actual delivery model. Override with the borrower's own headcount data where the credit team has it — and for a sponsor-backed borrower they usually do.

Discipline. Use the research to set the *severity* of a scenario and to challenge the credit team's segment scores — not to estimate a beta. The mapping from borrower to driver comes from the revenue segmentation, which you own; the research supplies the size of the shock, which you do not. And treat all of it as a lower bound on speed: published exposure work is a snapshot of a moving capability frontier.

14. Worked AI displacement scenario — S7 on Northwind

Same borrower, same formulas, no macro shock at all. Reference rate stays at 4.00%, spreads elsewhere are unchanged.

14.1 Step 1 — segment the revenue and score the exposure

Segment	Revenue	Share	Pricing basis	Dominant channel	Score (0–3)	3y gross displacement	Revenue at risk
Transactional BPO — AP/AR, claims processing	100.0	40%	per-FTE / per-transaction	1	3	45%	45.0
IT service desk, L1/L2	50.0	20%	per-seat / per-ticket	1	3	40%	20.0
Managed infrastructure	45.0	18%	fixed monthly	3	1	10%	4.5
Field and on-site services	35.0	14%	per-visit	—	0	0%	0.0
Regulated compliance outsourcing	20.0	8%	retainer, licensed staff	3	1	8%	1.6
Total	250.0	100%					71.1 = 28.4%

14.2 Steps 2–4 — the timing gate, the cost offset, and the cascade

Step	Computing	Formula	Substitution	Result	Test
1	Gross revenue at risk	$\Sigma \text{ segment} \times \text{displacement}$	from the table above	-28.4%	Over 3 years, before timing
2	Timing gate [13.3]	$\times \text{realisation within loan life}$	WA contract term 2.1y vs expected repayment 3.0y $\rightarrow$ 60% $-28.4\% \times 0.60$	-17.0%	Only what reprices before we are repaid
3	EBITDA, with cost offset [B1] + channel 3	$\Delta E\% = \Delta \text{Rev}\% \times \text{OL} + \text{cost deflation capture}$	$-17.0\% \times 2.00 = -34.1\%$ offset: 55% of COGS is offshore labour, borrower captures 30% $\rightarrow$ +6.1pp	-28.0% E_s = 36.0	Note the offset is real and must be modelled — ignoring it overstates the loss
4	Cash interest [B3]	$\text{Debt} \times (r_0 + m)$	$250 \times 9.50\% = \text{unchanged}$	23.75	No macro shock in this scenario
5	ICR [B4]	E_s / Int_s	$36.0 / 23.75$	1.516x	MARGINAL PASS — covenant 1.50x, headroom 1.1%
6	Leverage [B4]	$\text{NetDebt} / E_s$	$250.0 / 36.0$	6.94x	FAIL — covenant 6.50x
7	EV and moneyness [B5]	$\text{EV} = E_s \times \text{Mult}_s$ $\text{LTV} = \text{NetDebt} / \text{EV}$	$\text{Mult}_s = 11.0 \times 0.60 = 6.60x$ $\text{EV} = 36.0 \times 6.60 = 237.6$ $250.0 / 237.6$	105.2%	Loan underwater at going-concern value. Sponsor equity worthless
8	Recovery [B6]	$\text{EV}(1-d) - c \cdot \text{EV} - \text{senior}$	$237.6 \times 0.55 = 130.7$ less $0.08 \times 237.6 = 19.0$ less RCF 15.0 = 96.7	38.7% LGD 61.3%	d = 45%: asset-light and permanently impaired — nothing to wait for
9	PD and mark	$\Delta \text{DM} = 200 + 450$ $\Delta V/V = -\text{SD}_s \times \Delta \text{DM}$	PD 15.0% (1yr) $-3.50 \times 6.50\%$	-22.75%	Migration to CCC— on terminal-value impairment
10	Carry and the decision metric [H2, H3]	$\text{net accrual} - \text{PD} \times \text{LGD}$	$9.50\% \times (1 - 0.15 \times 0.5) = 8.79\%$ less $15.0\% \times 61.3\% = 9.20\%$	-0.41%	NEGATIVE. We are no longer being paid for the risk

14.3 The finding

	S3 — Stagflation (~1-in-25)	S7 — Technology displacement	Why they differ
Stressed EBITDA	36.25 (–27.5%)	36.00 (–28.0%)	Near-identical damage to the income statement, arrived at by completely different routes — one macro, one structural.
ICR	1.208× FAIL	1.516× marginal pass	
Leverage	6.90× FAIL	6.94× FAIL	
EV multiple applied	7.70× (cyclical –30%)	6.60× (structural –40%)	Cyclical multiples revert; structural de-ratings do not
LTV	89.6%	105.2%	The loan is underwater in S7 and merely stretched in S3
LGD	47.9%	61.3%	Higher distress discount: the assets are people and there is no recovery to wait for
Mark	–24.50%	–22.75%	Similar — the mark is the <i>least</i> informative comparison
Carry – expected loss [H3]	+5.53%	–0.41%	The decisive line. Opposite answers from near-identical EBITDA damage
Resolution path	Amend, extend, wait for the cycle. Multiple recovers, EBITDA recovers, we are repaid.	No cycle to wait for. Amend-and-extend just lengthens exposure to a shrinking business.	F5: we are repaid at the exit , and S7 impairs the exit permanently

Why this is worth building even though it carries no probability

Cyclical stress impairs cash flow temporarily; structural stress impairs the exit permanently. Direct lending is repaid by refinancing or sale (F5), so a structural scenario attacks the actual repayment mechanism rather than the interim coverage. That is why S7 produces a worse credit outcome than a 1-in-25 recession while doing the same damage to EBITDA — and why the standard toolkit, which watches ICR, would rate S7 as the milder of the two. **ICR is the wrong lens for structural risk. LTV and LGD are the right ones.**

And it defeats diversification. Displacement is a single common driver across every exposed name — Section 12.3(c). If 35% of the book sits in exposed services and software segments, the exposed cohort moves as one: 35% × 36% cohort loss ≈ **–12.6% of NAV with essentially no diversification benefit**, against a Vasicek model that would spread the same expected loss thinly across the book and report a far smaller tail. The portfolio effect is worse than the name-level effect suggests, which is the opposite of the usual intuition.

Applying the same method to climate transition risk requires no new machinery: it is also structural, also terminal-value-impairing, and also mapped through revenue segments rather than betas. Build S7 properly and the transition scenario is a re-parameterisation, not a new project.

15. MI pack, implementation and governance

15.1 The standing MI pack

One page monthly, quarterly deep-dive behind it. Every metric maps to the FSB's May 2026 private credit monitoring annex — worth stating in the cover note, since it converts an internal risk report into evidence of supervisory alignment.

Panel	Metrics	Why it earns its place
1. Distance to covenant	Distribution of EBITDA decline to breach (Section 8); exposure-weighted % within 10/20/30%; quarter-on-quarter migration; cure rights consumed	The leading indicator. Moves before losses.
2. Coverage and leverage	Weighted average ICR and net leverage; % of exposure with ICR < 1.5× and < 1.0×; % with FCCR < 1.0×; % with negative free cash flow after cash interest	FSB borrower-credit-quality metrics; the ECB's stated euro-area concern
3. Stress results	ΔNAV under S1–S8, both mark view and total-return view , split MTM / impairment / carry; the loss bridge	The headline with its decomposition, per Section 6 [H4]
4. Carry quality	Gross accrual, non-accrual adjustment, cash vs PIK split, excess carry (carry – EL) by cohort	The decision metric. Section 6 [H3]
5. Valuation integrity	Mark-to-comparable gap (A5); implied-DM dispersion (A6); NAV series autocorrelation ϕ ; average residual correlation (12.3)	Pre-empt the valuation-opacity question and detects emerging common factors
6. Distress indicators	PIK share of income and of loans; amendments and waivers in last 12 months; non-accruals; net rating migration; % with maintenance covenants	Direct lift from the FSB annex; industry-comparable
7. Concentration and vintage	Top-5/10 obligor share; top-3 sponsor share; sector share; N_eff (inverse Herfindahl); exposure by origination vintage	The exposure the models under-price
8. Liquidity and financing	NAV facility LTV vs covenant on the mark view; unfunded commitments vs liquid resources; distribution coverage; redemption coverage	The amplifier and the fund-level constraint
9. Reverse stress	Iso-loss curve with historical crisis points overlaid; NAV decline to financing covenant breach	The one chart that survives every challenge session
10. Emerging risk	Exposure-scored share of book by theme; timing gate (displacement vs maturity); excess carry for the exposed cohort	Sections 13–14, refreshed semi-annually

15.2 Implementation — where each engine lives

Do in FIS APT	Do in Excel	Do in neither — judgement, documented and owned by credit
Curve and spread factor scenarios consistently with the public books; historical replays from realised factor returns; cross-asset aggregation; FX and hedge legs; spread DV01 and factor attribution; the residual correlation diagnostic	Engine B in full (Section 5); carry (Section 6); portfolio roll-up (Section 7); reverse stress (Section 8); Vasicek distribution; distance-to-covenant; mark-to-comparable gap; the segmental AI method (Sections 13–14)	Scenario narratives; operating leverage; add-back reversal severity; segment exposure scores; distress discount d; LGD documentation add-ons; extension probabilities; migration severity

Non-negotiable APT configuration point

Represent every loan by its terms and conditions as a floating-rate note — reference rate and tenor, contractual margin with the PIK component flagged, floor, expected repayment date, current mark and implied DM, internal rating, sector, currency, seniority — so APT prices it cash-flow-analytically. **Do not allow returns-based beta estimation on private reval history for any sleeve.** If a sleeve must be estimated, de-smooth first and validate against the 0.6–0.9 range versus the leveraged loan index. This single configuration decision is worth more than any scenario design in this document.

15.3 Minimum viable build

The framework degrades gracefully. Everything in Sections 5 through 9 and 13 through 14 runs off one sheet, buildable in under two weeks:

```
Identity:      ID · borrower · sponsor · sector · vintage · currency · seniority · exposure · weight
Terms:        reference rate + tenor · margin (cash / PIK split) · floor · expected repayment
              · spread duration · current mark · implied DM · internal rating
              · hedge type and EXPIRY [B7]
Fundamentals:  revenue · REVENUE BY SEGMENT [13.1] · adjusted EBITDA · add-backs · operating leverage
              · net debt · gross debt · cash · capex · tax · entry EV multiple
              · covenant type and level · cure rights used / available [B8]
              · super-senior RCF: size, drawn, undrawn
              · WA contract duration [13.3] · documentation flags (cov-lite, transfer, baskets)
Derived:      ICR · leverage · FCCR · LTV · distance-to-covenant · liquidity runway
Shock block:  one column per Tier-1 sensitivity and per Tier-2 scenario
              → ΔEBITDA · ΔMark · ICR_s · breach flag · LGD_s · PD_s · ECL · carry · excess carry
```

Three fields are new in v2.0 and are the ones most likely to be missing today: **revenue by segment** (which unlocks Sections 13–14), **hedge expiry** (B7), and **cure rights consumed** (B8). All three sit in the credit files already; none is currently in a tape.

15.4 Governance and the honest caveats

- **Severity calibration.** State the intended return period for each Tier-2 scenario. A scenario without a stated severity cannot be challenged and therefore cannot be trusted. S7 and S8 deliberately carry *no* return period and must never be presented with one.
- **Backtesting.** Re-run S2 and S3 against realised GFC and COVID paths for a comparable book; where the model under-predicts, document the multiplier applied.
- **Beta hygiene register.** For every beta in use: source, sample, R^2 , crisis-window cross-check, prudence margin, last refresh. Any beta estimated on smoothed data is flagged as a lower bound.
- **Assumption ownership.** Operating leverage, add-back reversal, segment exposure scores, distress discounts and extension probabilities are *credit-team* judgements, not risk-team assumptions. Have them signed accordingly — it improves both the input and the reception of the output.
- **Known limitations, stated up front:** asset correlation ρ drives the entire tail of Engine C and is not observable for private credit; recovery assumptions are calibrated on an unusually benign period against an unusually creditor-hostile documentation trend; sponsor correlation is assumed rather than estimated; the emerging-risk scenarios carry no probability; and the whole framework is only as good as borrower financials that arrive one to two quarters late.

If only four things get built

1. **The distance-to-covenant surface (C5, Section 8).** Cheap, needs no model, leads realised losses by two to four quarters.
2. **The mark-to-comparable gap (A5).** One number for how much loss is already in the book but not yet in the marks.
3. **Excess carry — carry minus expected loss (Section 6 [H3]).** The single line that tells you whether you are being paid, and the only one that separated S7 from S3.
4. **The S3-versus-S2 rates-and-earnings pincer, plus the S3-versus-S7 comparison (14.3).** Together they make the two points that matter: the shock the factor model scores at zero is the one that breaks the covenants, and the shock that looks milder on ICR is the one that breaks the recovery.

Internal methodology proposal v2.0. All figures are illustrative and internally consistent within this document; calibration requires refresh from Bloomberg and licensed private credit data (Cliffwater CDLI, Lincoln Senior Debt Index, KBRA DLD, Morningstar/LSTA) at each jump-off before use in any submission or committee paper. Northwind Business Services Ltd is hypothetical. Scenarios are not forecasts, and S7 and S8 carry no probability by design. Market context drawn from the FSB *Report on Vulnerabilities in Private Credit* (6 May 2026) and the ECB *Financial Stability Review* special feature on private credit (May 2026). AI exposure sources: Anthropic Economic Index (O*NET task mapping, automation/augmentation split); OpenAI GDPval (44 occupations across 9 GDP sectors); Eloundou et al. occupational exposure scores; Felten AIOE; BLS OES occupation mix by industry. Companion document: *SWES Private Credit Framework v1.0*.